Optimization Models for Targeted Offers in Direct Marketing

Implementation and computational study of the exact formulations

after F. Talla Nobibon, R. Leus, F.C.R. Spieksma, *EJOR* 210 (2011) 670–683

Judecir Cavalcante
ist431224

PhD Programme in Engineering and Management
Department of Engineering and Management
Instituto Superior Técnico, Universidade de Lisboa

Optimization and Applications
Prof. José Rui Figueira
Homework 3

Problem and models follow [1]. **Blue** *marks the presenter's implementation, experiments and interpretation.*

Outline

1. Problem and data
2. Formulations
3. Implementation
4. Experimental design
5. Experiment 1: small instances
6. Experiment 2: groups S1 to S3
7. Summary

The problem

- A **direct-marketing campaign** offers products to selected customers. The paper addresses *retention* campaigns, which seek additional revenue from existing customers of a bank [1].
- Response models estimate the expected return $p_{ij} = r_{ij}\, DFV_{ij}$ of offering product j to customer i (acceptance probability times value to the firm); these estimates are **inputs** of the model.

Product targeting problem

Decide which products to offer to which customers so as to maximize the expected net profit of the campaign, subject to four business requirements:

1. a corporate **hurdle rate** R on the return on investment;
2. a **budget** B_j per product, because each business unit funds its own product;
3. a maximum number M_i of **offers per customer**;
4. a **minimum-quantity commitment** (MQC) O_j: a product used in the campaign is offered to at least O_j customers.

A **fixed cost** f_j is incurred whenever product j is used.

The problem is strongly NP-hard; even finding a solution with positive profit is NP-hard [1].

Implementation: Instance generation

Instances follow the rules of Section 5.1 of [1], with $C_j = \sum_i c_{ij}$ and $\bar{M} = \sum_i M_i / n$.

Parameter	Meaning	Rule
c_{ij}, p_{ij}	cost and expected return of an offer	integers in $\{1, 2, 3\}$ and $\{0, \ldots, 16\}$
R	hurdle rate	5%, 10% or 15%
M_i, variant s	few offers per customer	between 1 and $n/5$
M_i, variant ℓ	many offers per customer	between $\lceil n/3 \rceil$ and $\lceil 2n/3 \rceil$
O_j	minimum-quantity commitment	between $\bar{M}$ and $2\bar{M}$
B_j	budget: minimum, maximum or random	between $O_j C_j / m$ and $2\bar{M} C_j / m$
f_j	fixed cost	proportional to O_j and to the margin of product j

Implementation

- The generator follows these rules, with two small adjustments (rounding of O_j and range of the random budget) that keep the budgets consistent.
- Instances are reproducible from a fixed seed. Because the seed is the same, instances with the same n share their costs and returns.

Model M1

Variables: $x_{ij} = 1$ if product j is offered to customer i; $y_j = 1$ if product j is used in the campaign.

$$\max \sum_{i,j} (p_{ij} - c_{ij})\, x_{ij} - \sum_j f_j\, y_j \qquad (1)\ \text{net profit}$$

$$\text{s.t.} \sum_{i,j} p_{ij} x_{ij} \;\geq\; (1+R)\Big[\sum_{i,j} c_{ij} x_{ij} + \sum_j f_j y_j\Big] \qquad (2)\ \text{hurdle rate}$$

$$\sum_i c_{ij} x_{ij} \leq B_j, \qquad j = 1, \ldots, n \quad (3)\ \text{budget}$$

$$\sum_j x_{ij} \leq M_i, \qquad i = 1, \ldots, m \quad (4)\ \text{offers per customer}$$

$$\sum_i x_{ij} \leq m\, y_j, \qquad j = 1, \ldots, n \quad (5)\ \text{linking}$$

$$\sum_i x_{ij} \geq O_j\, y_j, \qquad j = 1, \ldots, n \quad (6)\ \text{MQC}$$

$$x_{ij},\ y_j \in \{0, 1\}. \qquad (7)$$

- (M1) has $mn + n$ binary variables and $1 + m + 3n$ constraints. Constraints (5) and (6) together state that a product is either unused or offered to at least O_j customers.
- In the LP relaxation, (5) $\sum_i x_{ij} \leq m\, y_j$ allows $y_j = \sum_i x_{ij}/m$, so the fixed cost f_j is charged at a rate of only f_j/m per offer. This weakens the LP bound of (M1).

Model M2

(M2) replaces the aggregated linking constraints (5) $\sum_i x_{ij} \leq m\, y_j$ by their **disaggregated** form

$$x_{ij} \leq y_j, \qquad i = 1, \ldots, m,\ j = 1, \ldots, n. \qquad (8)$$

Proposition 3 [1]

The convex hull of the feasible solutions of (M1) equals the convex hull of the solutions satisfying (2)–(4), (6) and (8), with $0 \leq y_j \leq 1$ and $x_{ij} \in \{0, 1\}$.

- Summing (8) over i gives (5); hence every LP solution of (M2) is an LP solution of (M1), and $z^{LP}_{(M2)} \leq z^{LP}_{(M1)}$.
- In the paper, (M2) also relaxes $y_j \in \{0, 1\}$ to $0 \leq y_j \leq 1$. It has $mn - n$ more constraints than (M1): mn constraints (8) replace the n constraints (5).
- In the paper, (M2) is the formulation solved with CPLEX 10.2.

Remark *(implementation)*

The implementation keeps $y_j \in \{0, 1\}$ in the integer version of (M2). By Proposition 3 this does not change the optimal value, and the LP relaxation is the same, since x and y are relaxed to $[0, 1]$ in both cases.

Model M3

For each product j, let $S_{pj}, p = 1, \ldots, k_j$, be the sets of customers with $|S_{pj}| \geq O_j$ and $\sum_{i \in S_{pj}} c_{ij} \leq B_j$. The binary variable z_{pj} equals 1 if product j is offered exactly to S_{pj}.

$$\max \sum_j \sum_p \left[\sum_{i \in S_{pj}} (p_{ij} - c_{ij}) - f_j \right] z_{pj} \tag{9}$$

$$\text{s.t.} \sum_j \sum_p \left[\sum_{i \in S_{pj}} \big(p_{ij} - (1+R)c_{ij}\big) - (1+R) f_j \right] z_{pj} \geq 0 \tag{10}$$ hurdle rate

$$\sum_j \sum_{p:\, i \in S_{pj}} z_{pj} \leq M_i, \qquad i = 1, \ldots, m \tag{11}$$ offers per customer

$$\sum_{p=1}^{k_j} z_{pj} \leq 1, \qquad j = 1, \ldots, n \tag{12}$$ one set per product

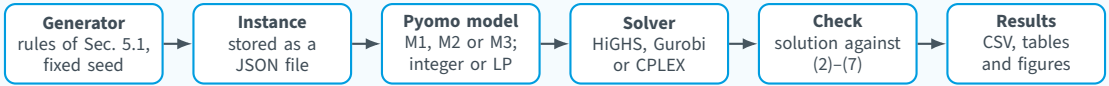
$$z_{pj} \in \{0, 1\}. \tag{13}$$

- Budget (3) and MQC (6) are built into the definition of the columns. (M3) results from a Dantzig–Wolfe decomposition, and $z^\star \leq z^{LP}_{(M3)} \leq z^{LP}_{(M2)}$ [1].
- The number of columns grows as $O(n\, 2^m)$. The paper solves (M3) by column generation within branch-and-price.

Remark *(implementation)*

The implementation enumerates all admissible columns explicitly and is therefore restricted to $m \leq 12$. Branch-and-price is not implemented. The solution of (M3) is mapped back to (x, y) and checked against (2)–(7).

Implementation: Overview



- **Pyomo as a common modelling layer.** Each formulation is written once, in algebraic form close to the paper, and sent to HiGHS, Gurobi or CPLEX by changing only the solver name.
- **M1 and M2 share one model builder**; they differ only in the linking constraints, (5) or (8). The LP relaxation changes the domain of the variables from $\{0, 1\}$ to $[0, 1]$.
- **M3 in two steps:** all admissible columns are enumerated, then the set-covering model is built from them.
- **Every solution is checked** against (2)–(7) by an independent function, and unit tests cover a small example solved by brute force.

Design: Experiments

Experiment 1 *(small instances)*

- 50 instances with $m = 4, \ldots, 12$ customers and $n = 2, \ldots, 5$ products.
- Models **M1, M2 and M3**.
- Goal: validate the three models and compare their LP bounds.

Experiment 2 *(groups S1, S2, S3 of the paper)*

- 162 instances with $m = 100$, 200, 300 customers and $n = 5$, 10, 15 products.
- Models **M1 and M2** (M3 needs column generation at this size).
- Goal: compare bounds and solve times as m grows.

In both experiments, each instance is solved as an integer program and as an LP relaxation, with **HiGHS, Gurobi and CPLEX**.

Computational environment

Machine	Intel Core i7-8565U @ 1.80 GHz (4 cores, 8 threads), 15.8 GB RAM, Windows 10
Software	Python 3.12.10, Pyomo 6.10.1
Solvers	HiGHS 1.15.1, Gurobi 13.0.3, CPLEX 22.2.0.0, with default settings
Time limit	300 s (Experiment 1) and 3600 s (Experiment 2) per run

Experiment 1: LP bounds

- The three models and the three solvers return the same optimum $z^\star$ on all 50 instances.
- The LP gap $100\,(z^{LP} - z^\star)/z^\star$ depends only on the model, and $z^\star \le z^{LP}_{(M3)} \le z^{LP}_{(M2)} \le z^{LP}_{(M1)}$ holds in every instance.

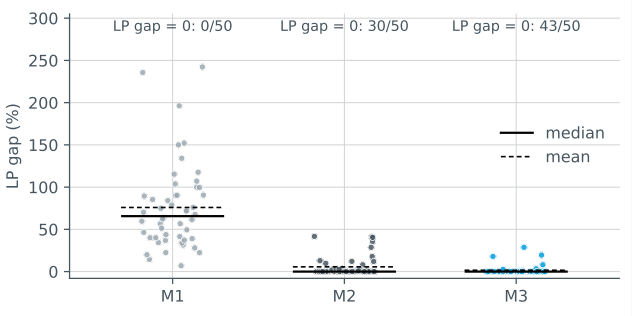


Figure 1. LP gap of M1, M2 and M3 on the 50 instances of Experiment 1; one point per instance.

	LP gap (%)			LP gap
Model	mean	median	max	$= 0$
M1	75.90	65.63	242.22	0/50
M2	5.63	0.00	41.67	30/50
M3	1.63	0.00	28.75	43/50

The LP relaxation of M3 already gives the integer optimum in 43 of 50 instances.

Experiment 1: Size of M3

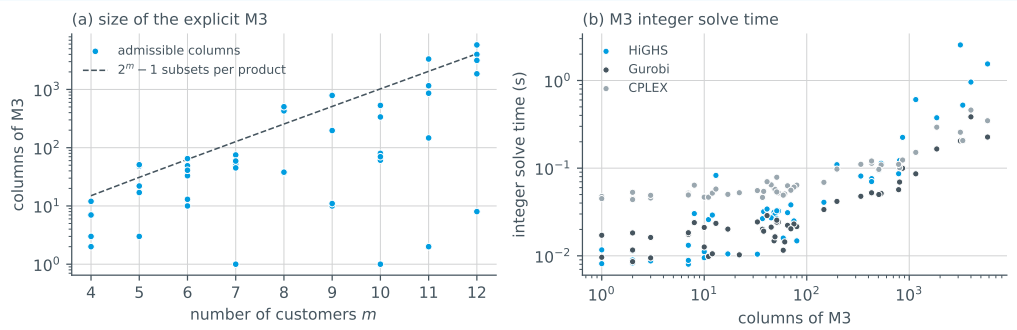


Figure 2. (a) Admissible columns of M3 per instance, as a function of m. (b) Integer solve time of M3 against its number of columns, per solver. Logarithmic scales.

- The number of columns ranges from 1 to 5 809, already with at most 12 customers.
- The solve time of M3 increases with the number of columns, but stays below 2.6 s.
- M1 and M2 are solved in less than 0.2 s on every instance; at this size the times are too small to rank the solvers.

Experiment 2: LP bounds

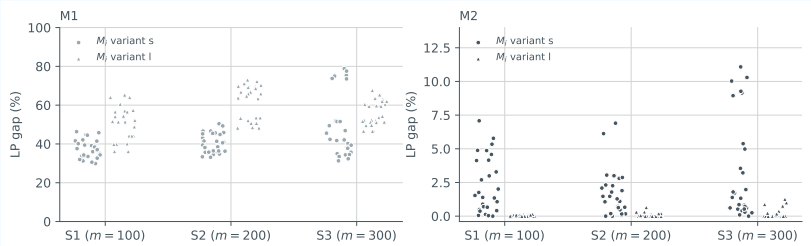


Figure 3. LP gap of M1 (left) and M2 (right) on the 162 instances, by group and M_i variant. The vertical scales differ.

Group	M1: LP gap (%) mean	M1: LP gap (%) max	M2: LP gap (%) mean	M2: LP gap (%) max
S1 ($m = 100$)	44.36	65.30	1.18	7.08
S2 ($m = 200$)	51.61	73.09	0.87	6.90
S3 ($m = 300$)	52.24	78.82	1.76	11.08

- All 972 integer runs end with status optimal and the models agree on the optimum; three runs on S3 stop at the solver tolerance (0.01%), one unit away from it.
- The LP gap of M2 is about 1% on average, against 44–52% for M1, in every group.

Experiment 2: Solve times

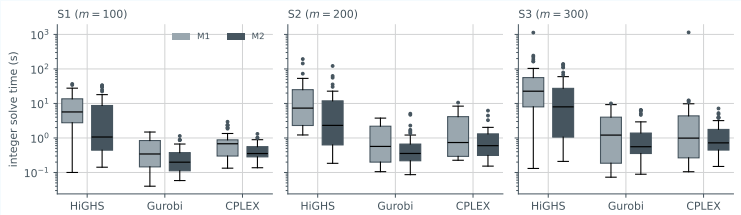


Figure 4. Integer solve time of M1 and M2 per solver, for each group (54 instances per box). Logarithmic scale.

Solver	Model	S1 ($m = 100$) median	S1 ($m = 100$) max	S2 ($m = 200$) median	S2 ($m = 200$) max	S3 ($m = 300$) median	S3 ($m = 300$) max
HiGHS	M1	5.68	36.71	7.33	193.25	22.65	1 128.02
	M2	1.07	33.85	2.32	122.02	7.97	139.14
Gurobi	M1	0.34	1.49	0.57	3.76	1.21	10.02
	M2	0.20	1.14	0.36	5.06	0.56	6.50
CPLEX	M1	0.68	2.95	0.74	10.59	0.99	1 145.38
	M2	0.35	1.31	0.60	6.21	0.72	7.15

- Gurobi and CPLEX solve a typical instance in about one second; HiGHS is roughly 7 to 9 times slower than Gurobi on M2.

Experiment 2: M1 versus M2

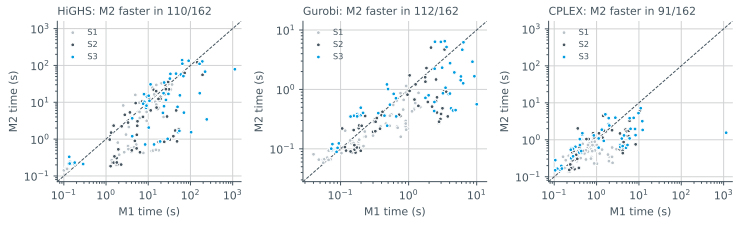


Figure 5. Integer solve time of M2 against M1 for each instance, per solver and group (logarithmic scales). Points below the diagonal are instances solved faster with M2.

	instances where M2 is faster			geometric mean of time(M2)/time(M1)		
Group	HiGHS	Gurobi	CPLEX	HiGHS	Gurobi	CPLEX
S1	35/54	44/54	36/54	0.46	0.66	0.69
S2	43/54	41/54	34/54	0.31	0.59	0.65
S3	32/54	27/54	21/54	0.44	0.71	0.75

- M2 is faster on average with every solver, but not on every instance.
- The time difference is much smaller than the difference in LP gap: the solvers probably strengthen M1 on their own (CPLEX solves 155 of the 162 M1 instances at the root node).

Experiment 2: Small versus large M_i

Variant s: each customer accepts few offers. Variant ℓ: each customer accepts many. Each variant has 81 instances.

	mean LP gap (%)		mean time of M2 (s)		
Variant	M1	M2	HiGHS	Gurobi	CPLEX
s (few offers)	42.05	2.43	24.74	1.14	1.35
ℓ (many offers)	56.76	0.10	3.05	0.37	0.55

- **Variant s is the harder case**: larger LP gap of M2 and solve times 2.5 to 8 times longer.
- **In variant ℓ, the LP relaxation of M2 is almost integral** (zero gap in 45 of 81 instances), and M2 gains most over M1.
- A possible explanation: when customers accept few offers, the products compete for the same customers.

Summary: Conclusions and limitations

Conclusions

- The three exact formulations of the paper were implemented and validated: all models and solvers agree on the optimum.
- The LP bounds follow the theory: M3 is the strongest and M1 by far the weakest. The disaggregated constraints (8) give most of the improvement.
- M2 is also faster on average and more robust than M1: its worst case is 139 s, against about 1 100 s for M1.
- Up to $m = 300$, M2 with a commercial solver is enough: at most 7.2 s per instance with Gurobi or CPLEX; HiGHS is several times slower.
- Instances where customers accept few offers are the harder ones.

Limitations

- M3 is solved only by enumeration ($m \leq 12$); column generation and branch-and-price are not implemented.
- Only the small groups (S1 to S3) were run, with one seed and one run per instance.
- The instances are generated by the rules of the paper, but they are not the paper's instances.

Thank you

Questions?

Judecir Cavalcante (ist431224)
PhD Programme in Engineering and Management
Department of Engineering and Management
Instituto Superior Técnico, Universidade de Lisboa

Optimization and Applications
Prof. José Rui Figueira
Homework 3

Reference

[1] F. Talla Nobibon, R. Leus, F.C.R. Spieksma. Optimization models for targeted offers in direct marketing: Exact and heuristic algorithms. *European Journal of Operational Research*, 210(3):670–683, 2011.